

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js/React front end, Firebase/GCP backend) for a provider-facing **SMART on FHIR** clinical decision support app, in a fast-paced early-stage environment.
- Extending a **Firebase Cloud Functions** backend that builds medication alternatives from **RxNorm** normalization, formulary and prior-authorization logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (**Python**, **FastAPI**, sentence-transformers, React, **Docker**), productionizing a published NLP method into a self-serve web tool with full ownership of architecture and technical decisions delegated by the PI.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and structured data-quality warnings; grew the backend suite to **40 tests** running hermetically in CI without ML dependencies via a deterministic fake embedding backend.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack platform that replaced manual spreadsheet workflows for the AI team, cutting manual work by **100+ hours/month** and lifting labeling throughput by **60%** (React, Node.js, MongoDB).
- Hardened backend reliability across the eTriage stack, adding **integration tests** and regression protection, hardening authentication and rate limiting, and remediating **20+ VAPT findings** against sensitive clinical data.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant eTriage service stable.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service (Python, FastAPI) with typed retrieval planning, deterministic extraction with LLM fallback, and cell-level provenance; hardened with **150+ automated tests** and deployed as a live service on **Google Cloud Run**.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** NYC TLC dataset via broadcast joins, Adaptive Query Execution, and shuffle-partition tuning; profiled stage-level timings in the Spark UI to localize bottlenecks.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built a Python agent that proposes queries and labels, scores them against multiple retrieval systems, and commits only accepted updates back to **Git** with report cards, guarded by schema checks, duplicate rejection, and read-only holdout protection for a reproducible, auditable workflow.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, Bash, SQL, C++, TypeScript, JavaScript, Java

Testing & CI: pytest, integration testing, automated testing, CI/CD, GitHub Actions, Git

Systems & Linux: Linux, Spark / PySpark, perf profiling, distributed systems, reproducible benchmarking

Backend & APIs: Node.js, Express, FastAPI, React, REST APIs, MongoDB, PostgreSQL

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes